Yaoyang Liu

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Education

The Hong Kong University of Science and Technology (HKUST) **Hong Kong**
PhD in Computer Science and Engineering *2025–Present*
- Research Focus: Interactive World Model, Video Generation, Computer Vision
- Advisor: Prof. Long Chen

University of Science and Technology of China (USTC) **Hefei, China**
Bachelor of Engineering in Artificial Intelligence *2021–2025*
- School of Information Science and Technology
- GPA: 4.13/4.3, Rank: 1/60

Research Experience

Advised by Prof. Jiebo Luo and Prof. Li Yuan **University of Rochester (Remote)**
Research on Multi-modal Large Language Models *2024*
- Participated in the ChronoMagic-Bench project as part of open-sora-plan
- Conducted data processing and benchmark development for metamorphic evaluation of text-to-time-lapse video generation
- Studied diffusion models and gained foundational knowledge in multi-modal large language models

Internship Experience

LightSpeed, Tencent **Shenzhen, China**
Research Internship, Tencent Project UP *2026.02–Present*
- Research on interactive world models and controllable video generation
- Built a complete training pipeline for interactive world models and adapted it to multiple interactive environments including Kart and CS2; currently extending it to the multi-player interactive setting
- Proposed Rate-Adaptive Rollout (RAR), which decouples generation speed from the training frame rate via dynamic rate switching, providing a practical approach to deploying interactive world models

Publications

- **Yaoyang Liu***,, Dongyu Yan*, Yihua Du, Shengju Qian, Zeke Xie, Xin Wang, Ying-Cong Chen, Long Chen. "RAR: Rate-Adaptive Rollout for Time-Consistent Interactive World Models" *Preprint*, 2026. (*Equal contribution)
- **Yaoyang Liu**, Yuechen Zhang, Wenbo Li, Yufei Zhao, Rui Liu, Long Chen. "SwiftI2V: Efficient High-Resolution Image-to-Video Generation via Conditional Segment-wise Generation." *Preprint*, 2026.
- Yuechen Zhang*, **Yaoyang Liu***, Bin Xia, Bohao Peng, Zexin Yan, Eric Lo, Jiaya Jia. "Magic Mirror: ID-Preserved Video Generation in Video Diffusion Transformers." *IEEE International Conference on Computer Vision (ICCV)*, 2025. (*Equal contribution)
- Yiming Zhao, Yu Zeng, Yukun Qi, **Yaoyang Liu**, Lin Chen, Zehui Chen, Xikun Bao, Jie Zhao, Feng Zhao. "V2P-Bench: Evaluating Video-Language Understanding with Visual Prompts for Better Human-Model Interaction." *International Conference on Learning Representations (ICLR)*, 2025.
- Shenghai Yuan, Jinfa Huang, Yongqi Xu, **Yaoyang Liu**, Shaofeng Zhang, Yujun Shi, Ruijie Zhu, Xinhua Cheng, Jiebo Luo, Li Yuan. "ChronoMagic-Bench: A Benchmark for Metamorphic Evaluation of Text-to-Time-lapse Video Generation." *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track (Spotlight)*, 2024.

Honors & Awards

HKUST RedBird PhD Award	2025
Hong Kong PhD Fellowship Scheme (HKPFS)	2025
USTC Guo Moruo Scholarship	2025
Undergraduate National Scholarship	2023
Artificial Intelligence Talent Class Scholarship	2023
Robogame Robot Competition, First Prize	2023
Undergraduate National Scholarship	2022

Technical Skills

ML/AI Frameworks: PyTorch, Diffusion Models, Transformers

Research Areas: Video Generation, Multi-modal Large Language Models, Computer Vision